

Project Initialization and Planning Phase

Date	06 July 2026
Team ID	xxxxxxx
Project Title	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to evaluate users' conceptual understanding through voice-based explanations using Artificial Intelligence. It combines speech-to-text transcription, semantic similarity analysis, audio feature extraction, and intelligent scoring to provide instant feedback on conceptual understanding and communication skills. The system improves learning efficiency by generating automated evaluation reports.

Project Overview	
Objective	To develop an AI-powered application that evaluates spoken conceptual explanations, analyzes communication quality, and provides accurate feedback with performance reports.
Scope	The project enables users to upload voice recordings, converts speech to text, evaluates conceptual understanding using AI models, analyzes speech fluency, and generates downloadable PDF reports through an interactive web application.
Problem Statement	
Description	Students and learners often lack immediate and objective feedback on their spoken explanations, making it difficult to assess conceptual understanding and communication skills.
Impact	The solution improves learning outcomes by providing instant AI-based evaluation, increasing confidence, enhancing communication skills, and reducing the need for manual assessment.
Proposed Solution	
Approach	Develop an AI-powered web application using Whisper for speech-to-text, Sentence-BERT for semantic analysis, Librosa for audio feature extraction, and Streamlit for an interactive user interface.

Key Features	• Voice recording/audio upload • Speech-to-text transcription • Semantic similarity analysis • Speech fluency evaluation • Waveform visualization • AI-generated feedback

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel i3/i5 or higher
Memory	RAM	Minimum 4 GB (8 GB Recommended)
Storage	Disk space	10 GB Free Space
Software		
Frameworks	Web Framework	Streamlit
Libraries	AI & Audio Libraries	Whisper, Sentence-BERT, Librosa, ReportLab, Matplotlib, SoundFile, NLTK
Development Environment	IDE	Visual Studio Code / PyCharm
Data		
Data	Audio Files & Reference Concepts	.wav, .mp3, Reference text dataset, SQLite Database

